

Solution Overview

Show a secure AWS-based AI recruitment and workforce scheduling platform from candidate input and HR workflows through AI processing, governance, analytics, and closed-loop feedback.

Functional Requirements

- bulk resume upload and ingestion
- interview audio upload with interviewer/candidate separation
- sentiment and keyword detection with interview summary generation
- weighted human-in-the-loop scoring with JD-to-resume matching
- closed-loop quality-of-hire feedback from probation outcomes
- explainable scheduling with conflict log and decision trace
- real-time cost and compliance impact analysis during schedule edits
- HR dashboard with recruitment and retention funnel analytics

Non-Functional Requirements

- high security and compliance controls
- role-based access and consent tracking
- resilient async processing for media and analytics workloads
- auditability and explainability for AI decisions
- near-real-time UI feedback for schedule changes
- scalable storage for resumes, audio, reports, and logs

Assumptions

- Use AWS managed services as the primary cloud since analysis detected AWS
- Represent model providers as managed AI endpoints rather than specific vendor internals
- Collapse repeated worker replicas into single labeled nodes when possible
- Show production path only; collapse non-production environments into summary references if needed

Context

Scale
medium
Deployment
aws
Security
high

Architecture Blueprint

Hybrid

The solution is hybrid because user-facing portal and decision APIs must stay synchronous, while audio analysis, resume parsing, report generation, and quality-of-hire feedback are better handled through event-driven workflows. Step Functions plus SQS/EventBridge decouple expensive AI work from the interactive UX, and Aurora PostgreSQL keeps the authoritative workflow state and audit trail. This also fits the elevated security and compliance needs by making consent, encryption, and traceability explicit.

Key Design Decisions

1. Public traffic lands on CloudFront and S3 for the portal, while API Gateway and Cognito isolate authenticated access to sensitive HR functions and consented interview data.
2. Resumes and interview recordings are stored in S3 and pushed into asynchronous workflows so large files never block the recruiter or candidate experience.
3. Textract, Transcribe, Comprehend, and Bedrock are staged as a controlled AI pipeline that produces structured evidence, summaries, and scores rather than exposing raw model output directly to end users.
4. Aurora PostgreSQL stores the system of record for consent, interview scores, scheduling state, and closed-loop feedback; Redis accelerates real-time schedule impact calculations; OpenSearch indexes evidence and conflict logs for explainability and retrieval.
5. CloudWatch, CloudTrail, and KMS enforce operational visibility, auditability, and data protection so the platform can satisfy Taiwan PDPA-style consent and governance requirements.
6. External probation/performance feedback is fed back into the platform through the event bus so quality-of-hire metrics can flag model bias and improve hiring and scheduling prompts over time.

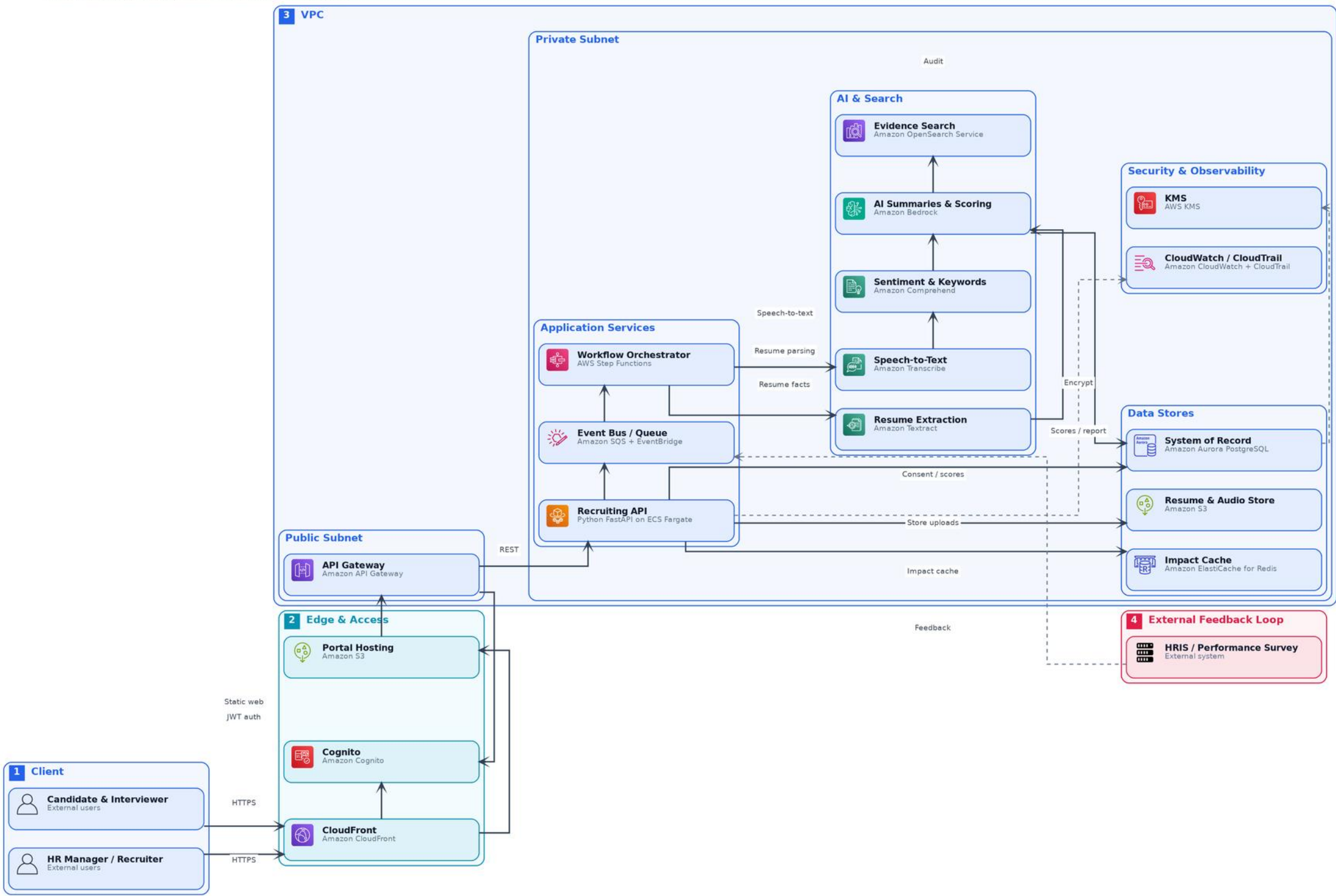
Tiers and Components

Client	Candidate & Interviewer, HR Manager / Recruiter
Edge & Access	CloudFront, Portal Hosting, Cognito, API Gateway
Application Services	Recruiting API, Workflow Orchestrator, Event Bus / Queue
AI & Search	Resume Extraction, Speech-to-Text, Sentiment & Keywords, AI Summaries & Scoring, Evidence Search
Data Stores	Resume & Audio Store, System of Record, Impact Cache
Security & Observability	KMS, CloudWatch / CloudTrail
External Feedback Loop	HRIS / Performance Survey

Architecture Diagram

AI Recruitment & Workforce Intelligence Platform

Compliant hiring, interview analytics, explainable scheduling, and closed-loop feedback



Production architecture for a compliant AI recruiting and scheduling platform